

Name **SUGGESTED SOLUTIONS**



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 1

Class Reg Number

17

Calculator Model / No.

Chemistry

8873/02

Paper 2

13 September 2018
2 hours

Additional Material: *Data Booklet*

READ THESE INSTRUCTIONS FIRST

Write your name, class and register number in the spaces at the top of this page.

This booklet contains Section **A** and Section **B**.

Section A (60 marks) : Page 2 to 15

Answer **all** questions in the spaces provided. You are advised to spend about **1 h 30 min** on Section **A**.

Section B (20 marks): Page 16 to 24

Answer **one** question in Section **B** in the spaces provided. You are advised to spend about **30 min** on Section **B**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

Examiner's Use		
Paper 1	MCQ	/ 30
	/ 33 %	
Paper 2 Section A	Q1	/ 15
	Q2	/ 17
	Q3	/ 16
	Q4	/ 12
Paper 2 Section B	/ 20	
Paper 2 Total	/ 80	
	/ 67 %	
Grand Total	/ 100 %	
Grade		

This document consists of **24** printed pages.

Section A

Answer **all** questions in this section, in the spaces provided.

- 1 Chemists have long studied the trend of elements, and the Periodic Table was celebrated as an effective way to arrange the elements while reflecting some of the most well-known trends in their physical and chemical properties.

- (a) Using structure and bonding, describe and explain the variation in melting points for elements aluminium to phosphorous.

[4]

The melting point increases from aluminium (Al) to silicon (Si), then decrease to phosphorous (P₄)

Aluminium: giant metallic lattice structure

Large amount of energy needed to overcome strong electrostatic forces of attraction between the cations and the sea of delocalised electrons.

Silicon: giant molecular structure

Larger amount of energy to overcome the strong and extensive covalent bonds between Si atoms in the giant 3-dimensional molecular structure.

Phosphorous: simple molecular structure

Small amount of energy to overcome the weak intermolecular instantaneous dipole-induced dipole forces of attraction between P₄ molecules.

- (b) (i) The ionisation energies of Period 3 elements from sodium to argon generally increases, with the exception of two decreases: from Mg to Al, and from P to S.

Explain these two anomalies in the trend.

[2]

From Mg to Al

Al has a lower 1st I.E as compared to Mg. The 3p electron in Al is further away from the nucleus compared to 3s electron in Mg. Hence, the 3p electron in Al experience weaker electrostatic attraction with the nucleus and hence require less energy to remove.

From P to S.

S has a lower 1st I.E as compared to P. Inter-electron repulsion is present in the doubly-filled 3p orbital of S atom. Hence, it is easier to remove the valence electron in the doubly-filled 3p orbital of S atom.

- (ii) An element from Period 3 was found to have the following successive ionisation energy values. Deduce the identity of this element.

Ionisation Energy	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th
Value / kJ mol^{-1}	790	1600	3200	4400	16000	20000	24000	29000

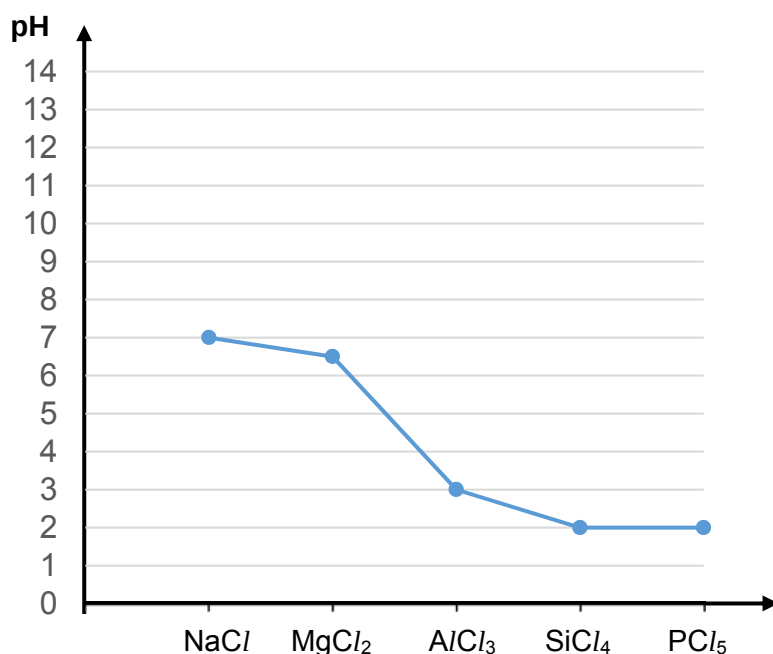
[1]

The element is silicon because there was a large increase from 4th to 5th ionisation energy. This indicates that the removal of the 5th electron was from the inner quantum shell and the element has 4 valence electrons.

- (c) Chlorides of Period 3 elements display a wide range of properties.

- (i) The pH of the resultant solution when Period 3 chlorides dissolves in or react with water shows a periodic trend. Sketch the pH trend from NaCl to PCl₅.

[1]



- (ii) Write an equation for the reaction of PCl₅ with water.

[1]



- (iii) One of the Period 3 chlorides and BeCl₂ have similar reaction with water. Identify this Period 3 chloride and explain why these two chlorides have similar chemical reactivity.

[1]

Period 3 chloride: AlCl₃

Be²⁺ and Al³⁺ have similar charge density

OR Be and Al have same atomic radius and electronegativity.

- (d) Describe and explain the relative reactivity of elements of Group 17 as oxidising agents.

[2]

Down the group,

- atomic radius increases.
- tendency for X_2 to accept electrons decreases.
- tendency for X_2 to be reduced to X^- decreases.
- oxidising power of X_2 decreases.

- (e) At the nanoscale, new properties may emerge in certain compounds.

For example, graphene is known to have very high tensile strength and high electrical conductivity. Describe the structure of graphene and explain how its structure relates to these two properties.

[3]

Graphene is a single layer of carbon atoms arranged in hexagons as they are in graphite (or a single layer of graphite). Each carbon atom in the layer is covalently bonded with 3 other carbon atoms resulting in a hexagonal lattice structure. Graphene has high tensile strength because of the strong covalent bonds between carbon atoms.

Each carbon atom has a 2p orbital that has a single electron that is not involved in bonding. This p orbital overlap sideways with its neighbouring carbon atom, resulting in an extended π -electron cloud of delocalised electrons above and below the plane. Graphene is an excellent conductor of electricity because it has delocalised mobile electrons over the layer which serves as charge carriers and can carry an electric current.

[Total: 15]

- 2(a)** Nearly all petrol and diesel vehicles have some kind of catalytic converter fitted to the exhaust to reduce the amount of pollutants emitted into the environment. The catalytic converter usually contains palladium nanoparticles which act as catalyst.

- (i)** Define the term nanoparticles.

[1]

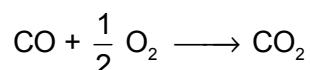
Nanoparticles are defined as particles with all dimensions between 1–100 nm on the nanoscale.

- (ii)** Explain why palladium used is in the form of nanoparticles instead of its bulk form.

[2]

Nanoparticles has high surface area per unit volume, i.e. the proportion of atoms on the surface of the nanoparticles is larger. As reaction occur on the surface of materials, chemical reaction often proceed more quickly OR rate of reaction increases.

- (iii)** The palladium catalyst helps to remove carbon monoxide from the exhaust. For example,



State the **type** of catalyst and outline the mode of action of catalyst in this reaction.

[2]

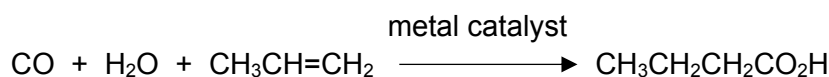
Heterogeneous catalyst

Adsorption: Reactant molecules are adsorbed onto the nanoparticle catalyst surface through the formation of *temporary bonds*.

Activation: This adsorption increases the surface concentration of the reactants and weakens the covalent bonds within the reactant molecules, thereby lowering the activation energy. Reactant molecules are brought closer together and reaction can take place between the reactants molecules more easily.

Desorption: Products diffuse away formed from the surface of the catalyst

- (b)** Carbon monoxide, water and propene can be used to produce butanoic acid according to the following equation:

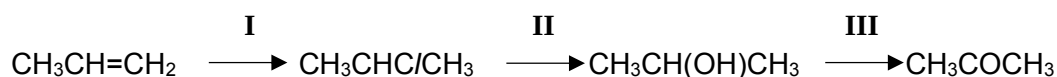


Draw the structure of the product formed when 2-methylpropene is used in the above reaction.

[1]



- (c) Propene can be converted to propanone by the following route.



- (i) Suggest the types of reaction for steps I and III.

[2]

	step I	step III
type of reaction	addition	oxidation

- (ii) State the reagent and conditions for step II.

[1]

aqueous NaOH, heat

- (iii) State and explain how the rate of reaction for step II changes when 2-chloropropane is replaced by 2-iodopropane.

[2]

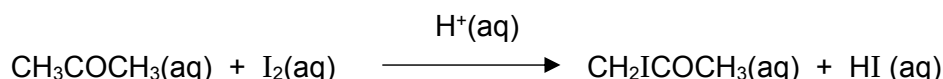
Relative reactivity of halogenoalkanes depends on the C-X bond strength.

C-I covalent bond is weaker than C-Cl

Relative ease of breaking C-I bond is greater than C-Cl

Hence rate of reaction increases when 2-chloropropane is replaced by 2-iodopropane.

- (d) The kinetics of the acid-catalysed reaction of propanone with iodine



can be investigated experimentally by varying the concentrations of the three substances involved and determining the time for the colour of the iodine to disappear.

In this method the rate of the reaction is measured in terms of the rate at which the iodine changes, i.e.

$$\text{rate} \propto \frac{\text{volume of aqueous iodine used}}{\text{time for the colour of iodine to disappear}}$$

The following results were obtained in such an investigation.

Expt	Volume of CH_3COCH_3 / cm^3	Volume of I_2 / cm^3	Volume of H^+ / cm^3	Volume of H_2O / cm^3	Relative time for the colour of iodine to disappear
1	8	4	8	0	1
2	8	4	4	4	2
3	4	4	8	4	2
4	8	2	8	2	0.5

- (i) Deduce the order of reaction with respect to propanone, iodine and hydrogen ions.

[3]

propanone

Comparing experiment 1 and 3, when conc. of propanone is halved while keeping the conc. of other reactants constant, rate is halved. Hence order of reaction w.r.t. propanone is 1

iodine

Comparing experiment 1 and 4, when conc. of iodine is halved while keeping the conc. of other reactants constant, rate does not change. Hence order of reaction w.r.t. iodine is 0.

hydrogen ions

Comparing experiment 1 and 2, when conc. of H^+ is halved while keeping the conc. of other reactants constant, rate is halved. Hence order of reaction w.r.t. H^+ is 1.

- (ii) Hence write the rate equation for this reaction.

[1]

$$\text{rate} = k [\text{CH}_3\text{COCH}_3] [\text{H}^+]$$

- (iii) Another experiment, with 0.80 mol dm^{-3} propanone, $0.0010 \text{ mol dm}^{-3}$ iodine and 0.20 mol dm^{-3} hydrochloric acid, is carried out at the same temperature.

Calculate the value of the rate constant if the rate is $4.2 \times 10^{-6} \text{ mol dm}^{-3} \text{ s}^{-1}$. Give the units of the rate constant.

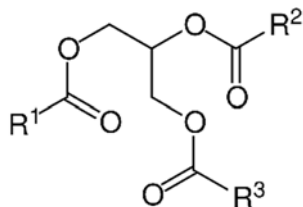
[2]

$$4.2 \times 10^{-6} = k(0.80)(0.2)$$

$$k = 2.6 \times 10^{-5} \quad \text{units: } \text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1}$$

[Total: 17]

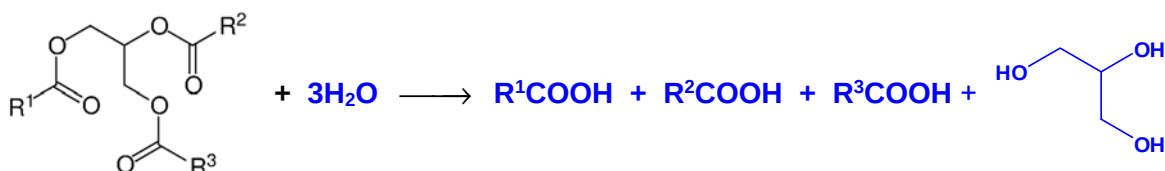
- 3(a)** Limited scientific evidence suggests that eating two tablespoons of olive oil daily may reduce the risk of coronary heart disease. Olive oil comprises mainly of the mixed triglyceride esters of oleic acid, palmitic acid and other fatty acids. The general chemical structure of olive oil is shown below where R^1 , R^2 and R^3 are alkyl groups or alkenyl groups.



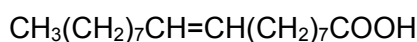
Olive oil undergoes rancidification when exposed to moisture, resulting in an unpleasant taste and odour. The products formed in the reaction are propane-1,2,3-triol and fatty acids.

Complete the equation to illustrate the rancidification reaction of olive oil.

[2]

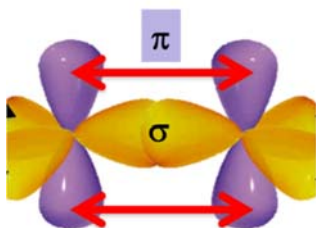


- (b)** Oleic acid has the structure shown.



- (i)** Describe the bonding in $\text{C}=\text{C}$ in terms of orbital overlap. Draw a diagram to illustrate your answer.

[2]



The $\text{C}=\text{C}$ double bond consists of 1 σ bond and 1 π bond.

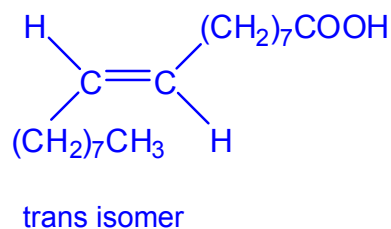
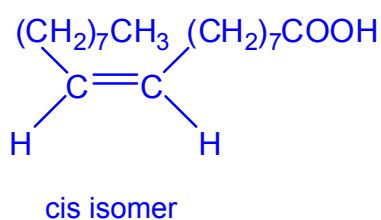
The $\text{C}-\text{C}$ σ bond is formed by the head-on overlap between one orbital of the carbon atom and one orbital of another carbon atom.

The $\text{C}-\text{C}$ π bond is formed by the sideways overlap between the $2p$ orbital of the carbon atom and $2p$ orbital of another carbon atom.

- (ii) Oleic acid exists as a pair of isomers.

Draw and label the structural formulae of the two isomers.

[2]



- (iii) Describe a chemical test that would distinguish oleic acid from ethanoic acid.

[2]

Add Br₂ in CCl₄ to each compound in the dark at r.t.p.

Oleic acid: Orange-red bromine solution is decolourised.

Ethanoic acid: Orange-red bromine solution is NOT decolourised.

- (c) Fig. 3.1 shows a bomb calorimeter, inside a controlled temperature water jacket, which is used to find an accurate value of standard enthalpy change of combustion of oleic acid.

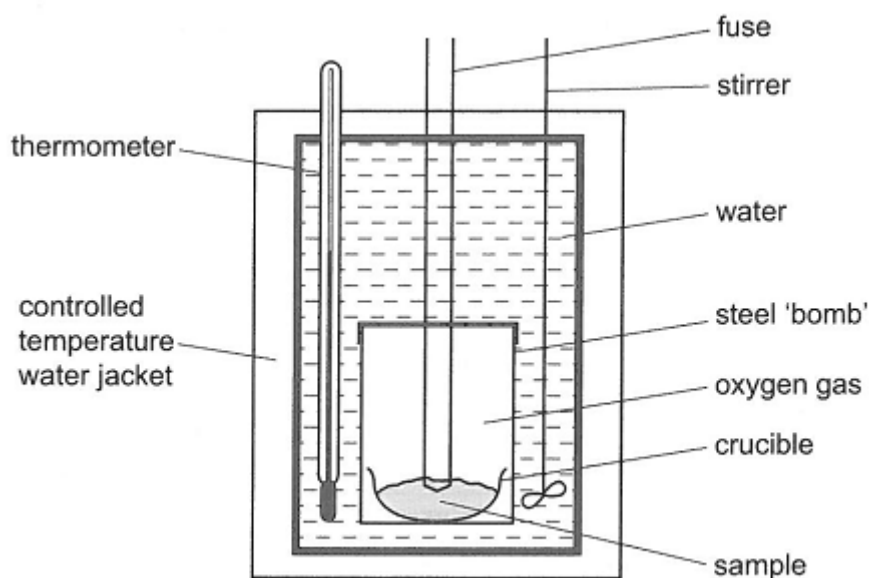


Fig. 3.1

- (i) What is meant by the *standard enthalpy change of combustion* of oleic acid?

[1]

It is the energy released when 1 mole of oleic acid is completely burned in (excess) oxygen under standard conditions (298 K and 1 bar).

- (ii) Explain why the steel 'bomb' is flushed with a high pressure of oxygen at the start of the experiment.

[1]

It is to ensure that there is sufficient O₂ present in the steel bomb for complete combustion.

- (iii) There is no heat lost from the calorimeter because of the controlled temperature water jacket. Suggest how the controlled temperature water jacket achieves this.

[1]

The temperature of water in the jacket is raised with an electric heater so that it just matches the average temperature of the calorimeter.

- (iv) A 2.98 g sample of oleic acid ($C_{18}H_{34}O_2$) is burned in a bomb calorimeter with a heat capacity of 1.98 kJ K^{-1} . The temperature of the calorimeter increases by 59.3°C .

Calculate the standard enthalpy change of combustion of oleic acid.

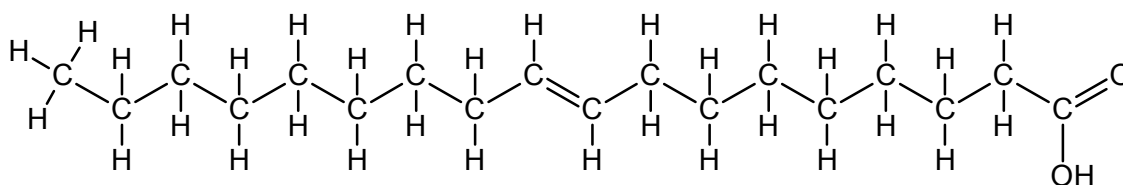
[2]

Heat absorbed by calorimeter = $1.98 \times 59.3 = 117 \text{ kJ}$

Amount of oleic acid used = $\frac{2.98}{282.0} = 0.0106 \text{ mol}$

$$\Delta H_c^\circ(\text{oleic acid}) = -\frac{117}{0.0106} = -11000 \text{ kJ mol}^{-1}$$

- (v) Use the bond energies given in the *Data Booklet* to calculate another value for the standard enthalpy change of combustion of oleic acid.

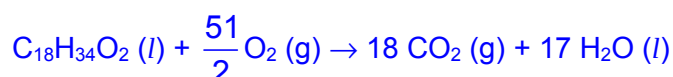


oleic acid

You may find the following information useful in your calculation.

An oleic acid ($C_{18}H_{34}O_2$) molecule contains 33 C–H bonds, 16 C–C bonds and other bonds.

[3]



Bond breaking (endothermic) → energy absorbed		Bonds forming (exothermic) → energy released	
33 C-H bonds	33 (410)	36 C=O bonds in CO_2	36 (805)
16 C-C bonds	16 (350)	34 O-H bonds	34 (460)
1 C=C bond	610		
1 C=O bond	740		
1 C-O bond	360		
1 O-H bond	460		
$\frac{51}{2}$ O=O bonds	$\frac{51}{2}$ (496)		
Total	33948 kJ mol^{-1}	Total	44620 kJ mol^{-1}

$$\Delta H_c^\circ(\text{oleic acid}) = 33948 - 44620 = -10672 = -10700 \text{ kJ mol}^{-1}$$

- (vi) Suggest a reason for the discrepancy between the value in (c)(iv) and that in (c)(v).

[1]

Bond energy values of polyatomic molecules in the *Data Booklet* are average values.

OR

Oleic acid is not in gaseous state.

[Total: 16]

4(a) There are two types of polymerisation, addition and condensation

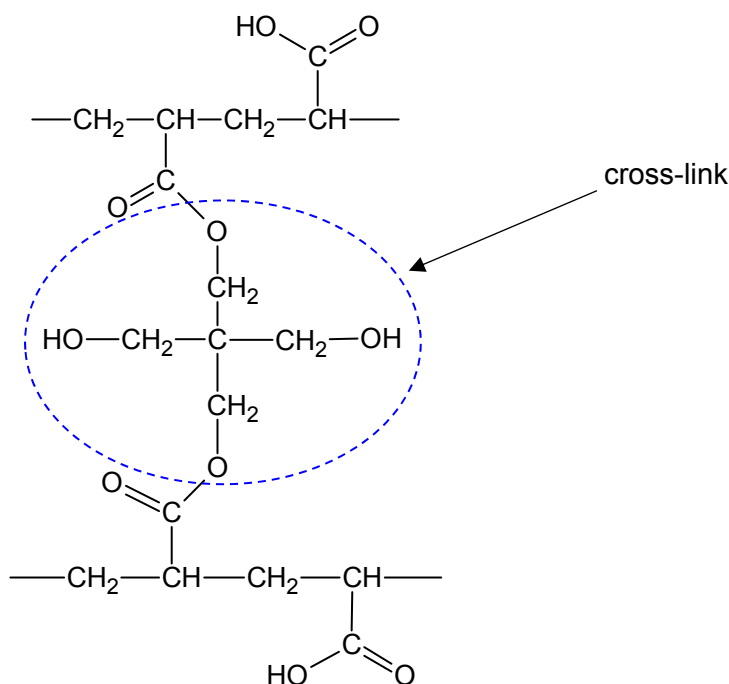
Account for the differences between these two types of polymerisation.

[2]

	Addition Polymerisation	Condensation Polymerisation
Type of monomers	Involves monomers containing double or triple bonds (unsaturated compounds)	Involves monomers containing alcohol, carboxylic acid or amino ($-\text{NH}_2$) functional groups.
By-products during polymerisation	No loss of molecules and the polymer is the only product.	Involves elimination of small molecules e.g. H_2O or HCl .
Empirical formula of polymer as compared to its monomers	Addition polymers have the same empirical formula as their monomers.	Empirical formula of the condensation polymer is not the same as that of the monomer.

(b) Super-absorbent polymers have the ability to absorb 200 – 300 times their own mass of water. They are classified as hydrogels and are widely used in personal disposable hygiene products such as diapers. However, these synthetically made hydrogels are usually non-biodegradable.

The diagram below shows part of the structure of a hydrogel.



The hydrogel is formed from chains of one polymer which are then cross-linked using another molecule.

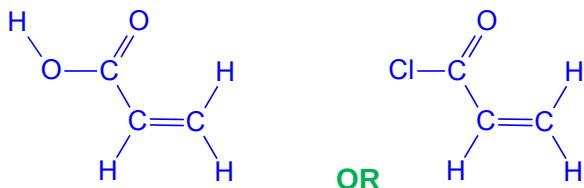
- (i) State the type of polymerisation used to form these polymer chains.

[1]

Addition polymerisation

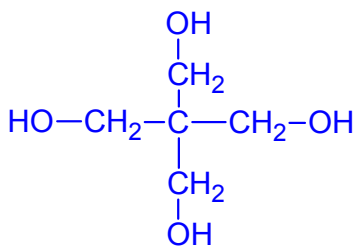
- (ii) Draw the **displayed** structural formula of the monomer used.

[1]



- (iii) Draw the structure of the molecule used to cross-link the polymer chains.

[1]



- (iv) Once a hydrogel has absorbed water, it can be dried and re-used many times. Explain why this is possible, referring to the structure.

[1]

Water is held to the hydrogel by hydrogen bonds formed between water molecules and COOH of the chain or OH of the cross-link. These hydrogen bonds can be easily broken to release the held water molecules during drying.

- (v) The amount of cross-linking has little effect on the ability of the gel to absorb water. Suggest why this is the case.

[1]

The amount of cross-linking does not affect the number of O-H groups available for hydrogen bonding.

- (v) Suggest one property of the hydrogel that will change if more cross-linking takes place. Explain how the increased cross-linking brings about this change.

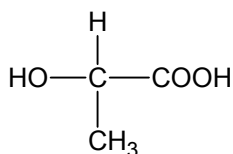
[2]

Hydrogel becomes harder / more rigid / less flexible / more brittle.

The strong cross-link covalent bonds between chains cannot be easily overcome. More cross-links hold the chains more tightly and strongly OR decreases the freedom of movement of individual chains

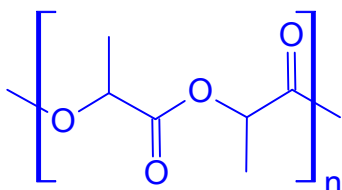
- (c) Unlike hydrogel, the polymer, 'polylactide' or PLA is biodegradable. The monomer required to produce PLA is *lactic acid*, 2-hydroxypropanoic acid.

The structure of *lactic acid* is as shown.



- (i) Draw the structure of the polymer PLA, showing two repeat units.

[1]



One of the reasons PLA has attracted so much attention is that it is biodegradable. This does, however restrict some potential uses. The polymer has a melting point of around 175 °C , but soften between 60 – 80°C. However, its properties enable it to have a range of uses in food packaging and in fibres.

- (ii) Explain why PLA would not be a suitable packaging materials for food pickled in vinegar.

[1]

PLA contains ester linkages in the polymer which can be hydrolysed by the acid present in vinegar e.g. CH₃COOH (aq), under prolonged exposure.

- (iii) Suggest why PLA containers are not usually used to hold hot drinks.

[1]

PLA is a thermoplastic polymer. PLA softens when heated and hence the PLA container changes shape easily as weak van der Waals' forces of attractions between chains are easily overcome and the chains are able to slide past each other.

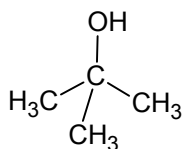
[Total: 12]

END OF SECTION A

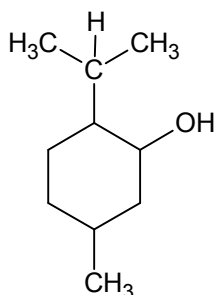
Section B

Answer **one** question from this section, in the spaces provided.

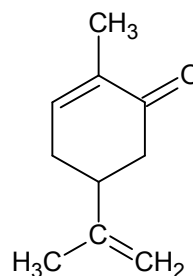
5(a) Three natural-occurring compounds used commercially are shown below:



2-methylpropan-2-ol



Menthol



Carvone

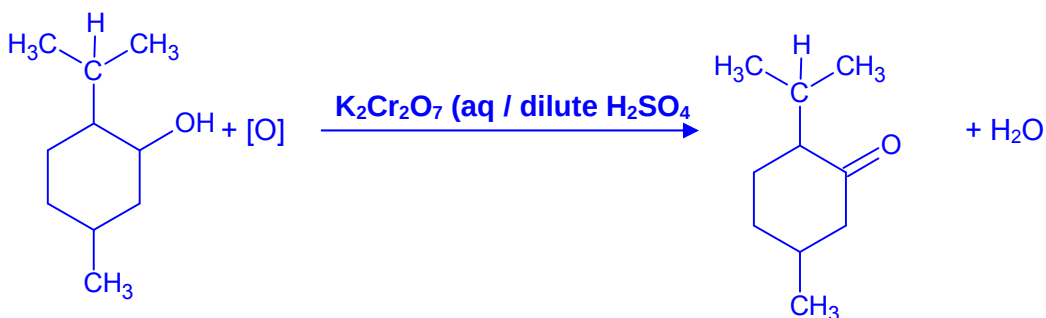
(i) Describe a set of two chemical tests that can be used to distinguish the above three compounds. Write an equation for each reaction that takes place.

[5]

Test: Add $\text{K}_2\text{Cr}_2\text{O}_7$ (aq), H_2SO_4 (aq) to each compound separately and heat.

Observations:

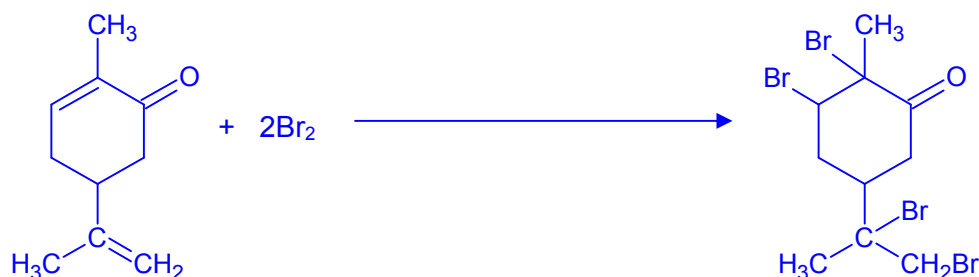
- 2-methylpropan-2-ol: Orange acidified $\text{K}_2\text{Cr}_2\text{O}_7$ remains.
- Menthol: Orange acidified $\text{K}_2\text{Cr}_2\text{O}_7$ turns green.
- Carvone: Orange acidified $\text{K}_2\text{Cr}_2\text{O}_7$ remains.



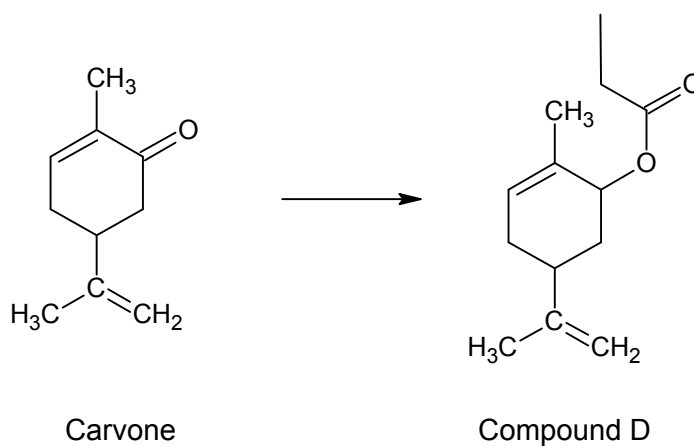
Test: Add Br_2 (CCl_4) / Br_2 (aq) to remaining compounds in the dark separately at r.t.p.

Observations:

- 2-methylpropan-2-ol: No decolourisation of orange-red Br_2 (CCl_4)
- Carvone: Decolourisation of orange-red Br_2 (CCl_4)

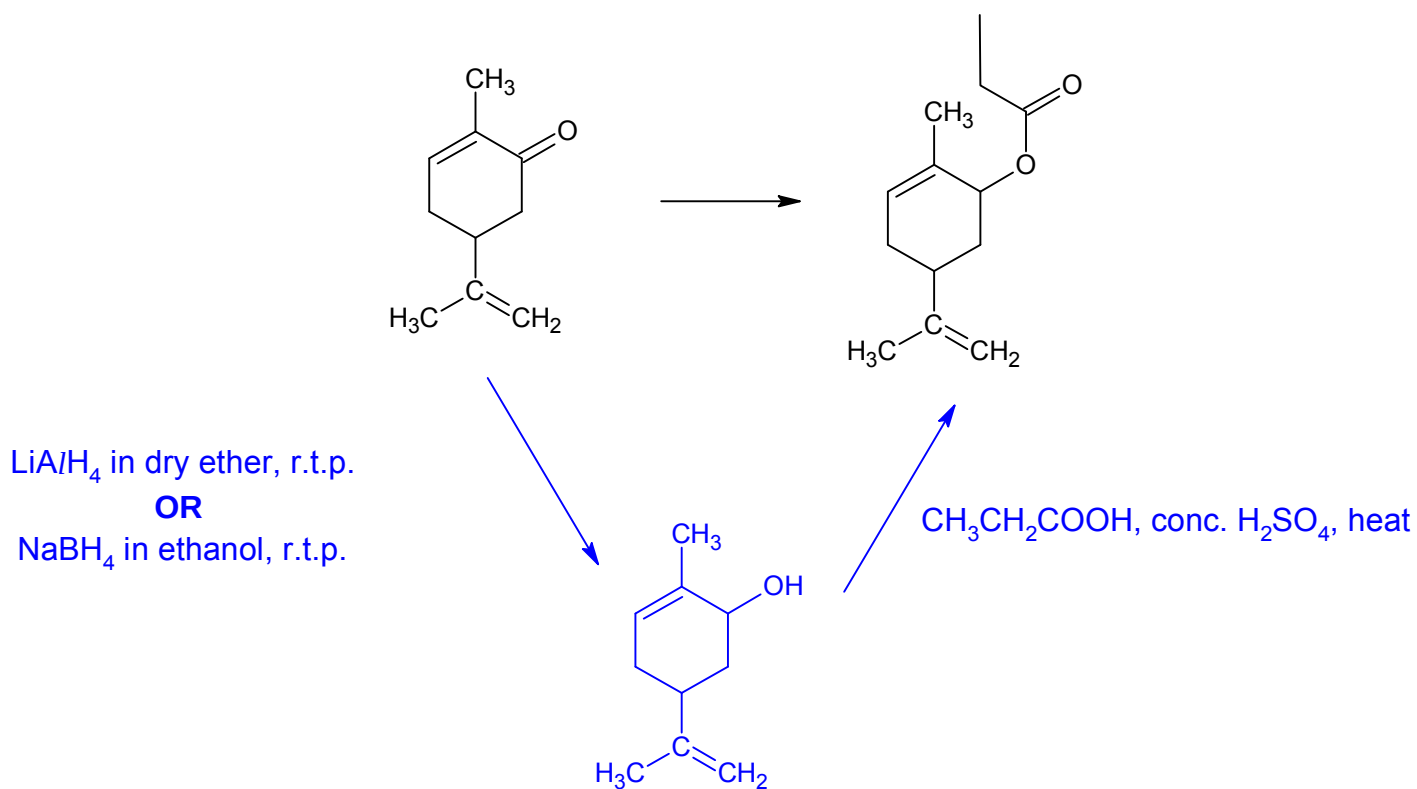


(ii) Carvone can be converted to organic compound D in two steps.



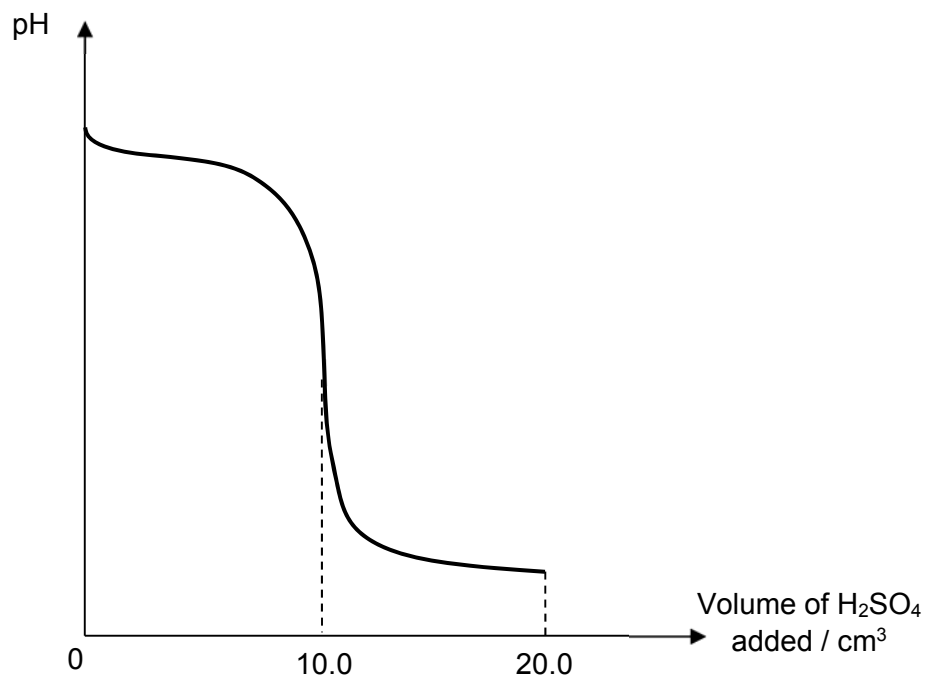
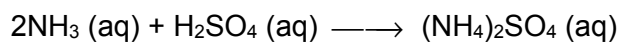
State the reagents and conditions necessary for each step. Give the structure of the intermediate involved in the conversion.

[3]



- (b) Ammonia was used in smelling salts to treat fainting. Modern smelling salt solution may contain other products such as lavender oil to act in conjunction with aqueous ammonia.

When 40 cm³ of aqueous ammonia was titrated against 0.200 mol dm⁻³ sulfuric acid, H₂SO₄ using a pH meter, the following graph was obtained.



The base dissociation constant for ammonia, K_b is $1.78 \times 10^{-5} \text{ mol dm}^{-3}$.

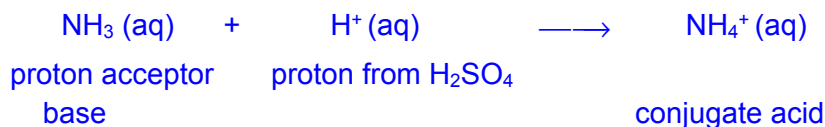
- (i) Explain what is meant by the terms *Bronsted–Lowry base* and *conjugate acid–base pair*. Illustrate your explanation using ammonia in the above reaction with sulfuric acid.

Explain why the *Arrhenius* definition of base does not apply for ammonia in the above reaction.

[3]

A Bronsted–Lowry base is a proton acceptor.

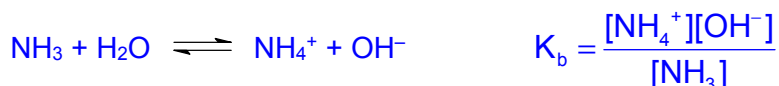
A conjugate acid–base pair is a pair of species that differs from each other by a proton.



An Arrhenius base is a substance that has OH in its formula and releases OH⁻ when dissolved in water. This definition does not explain basic property of ammonia that does not contain the OH group.

- (ii) Write an expression for the base dissociation constant, K_b , of ammonia.

[1]



- (iii) Using the titration curve provided, state the volume of sulfuric acid needed for the exact neutralisation of 40 cm³ of aqueous ammonia. Calculate the initial concentration of aqueous ammonia.

[2]

From the graph, the equivalence volume is 10.0 cm³.

$$\text{Amount of H}_2\text{SO}_4 \text{ in } 10.0 \text{ cm}^3 = \frac{10}{1000} \times 0.200 = 0.00200 \text{ mol}$$

Mole ratio: $2\text{NH}_3 \equiv \text{H}_2\text{SO}_4$

Amount of NH_3 in 40.0 cm³ solution = 0.00400 mol

$$\text{Initial concentration of ammonia in } 40.0 \text{ cm}^3 = \frac{0.00400}{\left(\frac{40.0}{1000}\right)} = 0.100 \text{ mol dm}^{-3}$$

- (iv) The resultant solution at the end of the titration can be considered as a sulfuric acid solution of concentration 0.0333 mol dm⁻³. Calculate the pH of this resultant solution to 2 decimal place.

[1]

$$[\text{H}_2\text{SO}_4] = 0.0333 \text{ mol dm}^{-3}$$

Since H_2SO_4 is a strong diprotic acid,

$$[\text{H}^+] = 2 \times 0.0333 = 0.0666 \text{ mol dm}^{-3}$$

$$\text{pH} = -\lg [\text{H}^+] = -\lg (0.0666) = 1.18 \text{ (2 d.p.)}$$

- (v) Suggest an appropriate indicator for the titration, giving a reason for your choice.

[1]

Methyl orange / methyl red may be used for this strong acid–weak base titration.

The pH transition range of the indicator (around 3.0 – 4.5) lies within the region of the rapid pH change (around 3 – 7) over the equivalence point.

- (vi) Explain what do you understand by the term *buffer solution*.

[1]

A buffer solution is one which is capable of maintaining a fairly constant pH when small amounts of acid or base are added to it.

- (vii) The *Henderson–Hasselbalch* equation is useful for the estimation of the pH of a buffer solution.

An alternative form of the equation is shown below which can be used to determine the pOH of a basic buffer solution.

$$\text{pOH} = \text{p}K_b + \log_{10} \frac{[\text{BH}^+]}{[\text{B}]}$$

Calculate the pH of a basic buffer solution, if the solution contains NH_4^+ and NH_3 in a 2 : 1 ratio. given that the base dissociation constant, K_b of ammonia, NH_3 is $1.78 \times 10^{-5} \text{ mol dm}^{-3}$.

[2]

$$\text{pOH} = \text{p}K_b + \log_{10} \frac{[\text{NH}_4^+]}{[\text{NH}_3]}$$

$$\text{pOH} = -\log (1.78 \times 10^{-5}) + \log_{10} \left(\frac{2}{1} \right)$$

$$\text{pOH} = 5.05$$

$$\begin{aligned} \text{pOH} &= 14 - \text{pH} &= 14 - 5.05 \\ & &= 8.95 \end{aligned}$$

- (viii) The enthalpy changes of neutralisation involving sulfuric acid with two bases are shown below.

sodium hydroxide, NaOH	– 57.3 kJ mol ^{–1}
ethylamine, CH ₃ CH ₂ NH ₂	– 51.5 kJ mol ^{–1}

Comment and explain for the above observation as fully as you can.

[1]

Ethylamine is a weak base. CH₃CH₂NH₂ is only slightly dissociated in aqueous solution.



Some of the energy evolved from the neutralisation process is used to further dissociate the weak base completely.

Hence, the enthalpy change of neutralisation involving CH₃CH₂NH₂ is less exothermic.

[Total: 20]

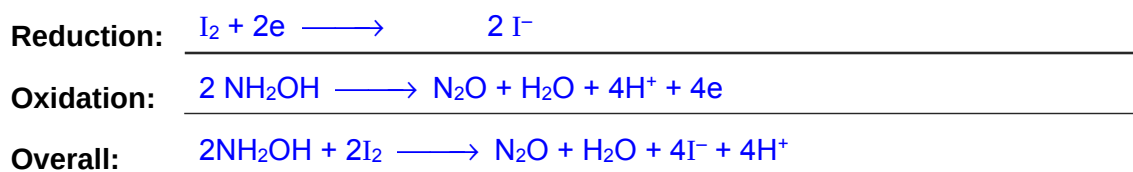
- 6(a)** Hydroxylamine, NH_2OH is commonly used in photography developing solution. Prolonged exposure to hydroxylamine by photography developers might cause irritation to their respiratory tract. To minimise the health hazard on these developers, the safety limit of the concentration of hydroxylamine is capped at below $0.00200 \text{ mol dm}^{-3}$.

A scientist performed the following experiment to determine if a sample of photography developing solution was within the safety limit.

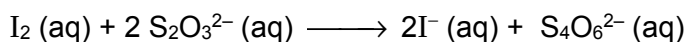
100 cm^3 of NH_2OH solution was added to 40 cm^3 of $0.015 \text{ mol dm}^{-3}$ acidified I_2 (aq) solution. In this reaction, iodine is used in excess and NH_2OH is oxidised to form dinitrogen monoxide, N_2O .

- (i) Using half-equations, write a balanced equation for the reaction between NH_2OH (aq) and I_2 (aq) in an acidic medium.

[2]



A 25.0 cm^3 aliquot of the resultant reaction mixture was then titrated with a standard solution of $0.00500 \text{ mol dm}^{-3}$ sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$ (aq).



Starch solution was added as indicator for this titration to enhance the contrast of the colour change at the end-point of the titration. 22.50 cm^3 of $\text{Na}_2\text{S}_2\text{O}_3$ (aq) was required.

- (ii) State the expected colour change at the end-point of the titration.

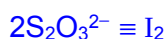
[1]

Blue-black to colourless.

- (iii) Calculate the amount of I_2 remaining in the reaction mixture.

[2]

$$\text{Amount of } \text{Na}_2\text{S}_2\text{O}_3 \text{ required} = \frac{22.50}{1000} \times 0.00500 = 1.125 \times 10^{-4} \text{ mol}$$



$$\begin{aligned} \text{Amount of } \text{I}_2 \text{ remaining in } 25 \text{ cm}^3 \text{ of the reaction mixture} &= \frac{1}{2} \times 1.125 \times 10^{-4} \\ &= 5.625 \times 10^{-5} \text{ mol} \end{aligned}$$

$$\begin{aligned} \text{Amount of } \text{I}_2 \text{ remaining in reaction mixture (140 cm}^3\text{)} &= \frac{140}{25.0} \times 5.625 \times 10^{-5} \\ &= 3.15 \times 10^{-4} \text{ mol} \end{aligned}$$

- (iv) Calculate the concentration of NH_2OH in 100 cm^3 sample of the photography developing solution. Hence conclude whether the solution is within the safety limit.

[2]

$$\begin{aligned}\text{Amount of } \text{I}_2 \text{ reacted with } \text{NH}_2\text{OH} &= \left(\frac{40.0}{1000} \times 0.015 \right) \text{ mol} = 3.15 \times 10^{-4} \text{ mol} \\ &= 2.85 \times 10^{-4} \text{ mol}\end{aligned}$$



$$\text{Amount of } \text{NH}_2\text{OH} \text{ present in } 100\text{ cm}^3 = 2.85 \times 10^{-4} \text{ mol}$$

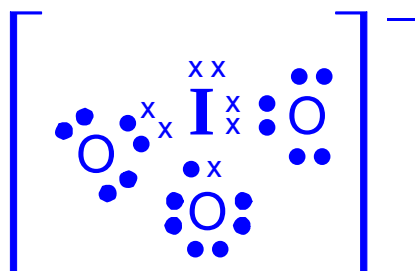
$$[\text{NH}_2\text{OH}] = 2.85 \times 10^{-4} \text{ mol} / (100/1000) = 0.00285 \text{ mol dm}^{-3}$$

Solution is $> 0.00200 \text{ mol dm}^{-3}$, hence it is NOT within the safety limit.

- (v) Iodate(V) ion, IO_3^- was used to generate iodine required for the reaction with NH_2OH .

In this IO_3^- ion, the central iodine atom can expand its octet structure. Draw the dot-and-cross diagram of IO_3^- ion.

[1]



- (vi) From your answer in (a)(v), state the bond angle around the central iodine atom and account for the shape of IO_3^- ion.

[2]

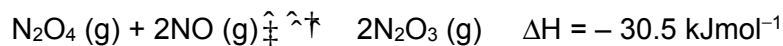
IO_3^- have 3 bond pairs and 1 lone pair around the central I atom.

To minimise repulsion and maximise stability, the 4 electron pairs are directed toward the corners of a regular tetrahedron.

Lone pair – bond pair repulsion $>$ bond pair – bond pair repulsion

IO_3^- is trigonal pyramidal, with a bond angle of 107°

- (b) In an industrial process, 5 mol of dinitrogen tetroxide, N_2O_4 and 7 mol of nitrogen monoxide, NO was heated in a 2.0 dm^3 vessel. The temperature is kept at 500°C . The two gases react slowly to form blue dinitrogen trioxide, N_2O_3 according to the following equation. It was found that the equilibrium mixture contains 2.3 mol of NO .



- (i) Write an expression for the equilibrium constant, K_c for the reaction, stating the units.

[1]

$$K_c = \frac{[\text{N}_2\text{O}_3]^2}{[\text{N}_2\text{O}_4][\text{NO}]^2} \text{ mol}^{-1} \text{ dm}^3$$

- (ii) Determine the equilibrium concentrations of the three gases. Hence, calculate the value of K_c .

[2]

	$\text{N}_2\text{O}_4 (\text{g})$	$+ 2\text{NO} (\text{g})$	$\rightleftharpoons 2\text{N}_2\text{O}_3 (\text{g})$
Initial amount / mol	5	7	0
Change in amount/ mol	$-\frac{1}{2}(4.7) = -2.35$	- 4.7	+4.7
Equilibrium amount / mol	2.65	2.3	+4.7

$$K_c = \frac{[\text{N}_2\text{O}_3]^2}{[\text{N}_2\text{O}_4][\text{NO}]^2}$$

$$K_c = \frac{\left(\frac{4.70}{2.0}\right)^2}{\left(\frac{2.65}{2.0}\right)\left(\frac{2.30}{2.0}\right)^2}$$

$$K_c = 3.15 \text{ mol}^{-1} \text{ dm}^3$$

- (iii) Predict and explain the effect of increasing temperature on the equilibrium position and equilibrium composition.

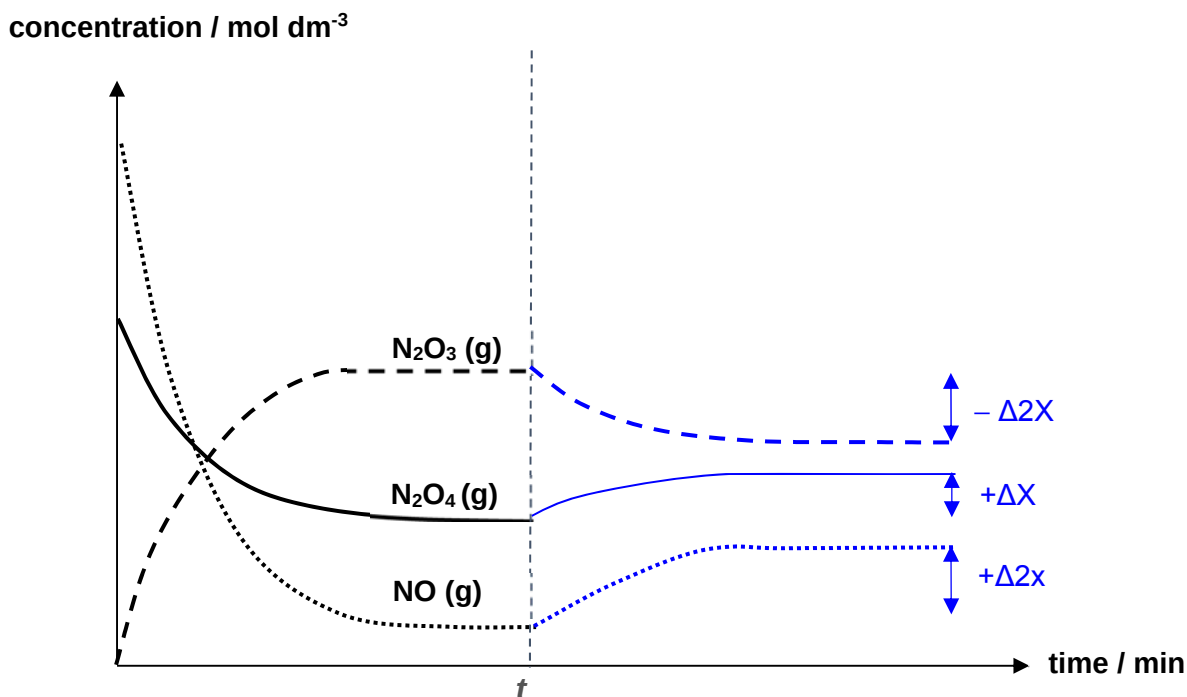
[2]

As temperature increases, the system will try to remove the heat added. By Le Chatelier's Principle, the equilibrium position will shift left towards the endothermic reaction to absorb heat.

The new equilibrium mixture contains less product, N_2O_3 and more reactants N_2O_4 and NO .

- (iv) Sketch on the same axes, how the concentration of the three gases will change when temperature was increased at t .

[2]



- (v) State the operating pressure and temperature (high or low) required to obtain a high yield of N_2O_3 . Explain your answer.

[2]

To obtain a high yield of N_2O_3 , a high pressure and low temperature should be used instead.

At high pressure, the system will try to reduce the pressure

By Le Chatelier's Principle, the equilibrium position shifts right to decrease the number of moles of gases to decrease pressure. $[\text{N}_2\text{O}_3]$ increases, resulting in a higher yield.

To obtain a high yield of N_2O_3 , a low temperature should be used instead.

At low temperature, the system will try to produce heat. By Le Chatelier's Principle, the equilibrium position will shift right towards the exothermic reaction to release heat. $[\text{N}_2\text{O}_3]$ increases, resulting in a higher yield.

- (vi) Hence, suggest why the operating temperature is fixed at a moderately high temperature of 500 °C.

[1]

If the temperature is too low, the rate of reaction will be too slow. Hence, a moderately high temperature is chosen as a compromise between the conflicting demands of rate and yield of the reaction.

[Total: 20]

END OF PAPER